

# Louay Guetat

## FULL STACK ENGINEER

Tunis, Tunisia · Open to relocation worldwide

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*Full Stack Engineer with 2+ years shipping production software across web, mobile, and cloud. I currently build fintech products end to end for a US startup — from React and React Native front ends to Python and Node.js APIs on AWS — with AI/LLM features throughout. Comfortable owning the whole stack: product UI, serverless backends, data pipelines, and CI/CD.*

## WORK EXPERIENCE

### Full Stack Engineer — Alpha Momentum

Remote · US Startup | February 2025 – Present

Financial research platform with factor models, SEC filing search, research automation, and live trading through Alpaca. I own most of the product stack, from the React web and mobile clients to Python and Node.js services on AWS. Since August 2026 I have also led Capital Thread, an AI diligence platform for private markets (detailed under Selected Projects), as a client engagement through Alpha Momentum.

- Built the React 18 + TypeScript web app (Vite, Material-UI, Tailwind): AWS Cognito/OAuth authentication, real-time SSE streaming for AI responses, a SEC document browser, interactive dashboards with Recharts, Stripe billing, and Excel export
- Built a serverless Node.js API (Express, Serverless Framework) on AWS Lambda for auth, portfolios, billing, and query routing, integrating Plaid (banking/webhooks), Alpaca (OAuth trading), Stripe, and AWS services (S3, SES, Cognito) over MongoDB/Mongoose
- Built Python FastAPI services: a SEC filing pipeline processing 37k+ documents on AWS Batch, data scrapers (BeautifulSoup/Selenium/aiohttp) for FMP/Fed/ECB/news feeds, and document search backed by Neo4j
- Shipped cross-platform React Native/Expo apps for iOS and Android with Plaid portfolio sync, Alpaca trading, in-app purchases (react-native-iap), and Firebase Analytics
- Set up LangChain and LangGraph workflows for equity research, valuation, portfolio optimization, macro, and risk, plus a RAG pipeline with OpenAI embeddings and Neo4j
- Built CI/CD on GitHub Actions across dev, staging, and production: Docker builds to ECR, ECS redeploys, S3/CloudFront releases, and Serverless Framework infrastructure-as-code

**Tech Stack:** TypeScript, Python, React 18, React Native, Node.js (Express, Serverless), FastAPI, AWS (Lambda, ECS, ECR, S3, CloudFront, Cognito, DynamoDB, Batch), Docker, MongoDB, Neo4j, PostgreSQL, LangChain, LangGraph, Stripe, Plaid, Alpaca

### Full Stack Developer — WeTekup

Tunis | March 2025 – September 2025

- Designed and built responsive, mobile-first React websites with a component-based architecture and SCSS (variables, mixins, modular patterns)

- Delivered robust REST APIs with Node.js/Express — auth, validation, and error-handling middleware — over normalized PostgreSQL schemas with optimized queries and indexing
- Deployed to production on Hostinger with Nginx as a reverse proxy: SSL/TLS, gzip, caching, and performance tuning (lazy loading, code splitting, asset optimization)

**Technologies:** React, Node.js, Express.js, SCSS, PostgreSQL, Nginx, REST API

### Data Science Intern — Sagemcom

*Tunis | February 2024 – August 2024*

- Built an end-to-end data pipeline from collection to model integration, scraping and assembling datasets with Selenium and BeautifulSoup
- Built retrieval pipelines with HuggingFace models and LangChain, plus automated workflows with LangGraph
- Developed a ReactJS + Flask web interface with REST API integration for model interactions

**Technologies:** Python, Flask, ReactJS, LangChain, LangGraph, HuggingFace, SQLite, REST API

### Full Stack Intern — WeTekup (Khedma.site)

*Tunis | July 2023 – September 2023*

- Built a job-marketplace platform for employers and job seekers with a ReactJS front end
- Designed and implemented REST APIs with Django over an optimized PostgreSQL schema

**Technologies:** ReactJS, Django, Python, PostgreSQL, REST API

### Frontend Developer Intern — E-Build

*Tunis | July 2022 – September 2022*

- Built and maintained responsive frontend components for web applications

**Technologies:** HTML, CSS, JavaScript

### Software Developer Intern — OACA

*Tunis | February 2021 – June 2021*

- Designed and developed a Java (JavaX) desktop application for market management and monitoring
- Produced UML diagrams and architecture documentation

**Technologies:** Java, JavaX, UML

## SELECTED PROJECTS

### Capital Thread — AI Deal Platform for Private Markets · Client engagement via Alpha Momentum · Aug 2026 – Present

AI diligence platform for private credit, private equity, and venture funds. Sources opportunities from SEC EDGAR and Form D bulk data, then runs a one-click diligence pipeline that parses Inline XBRL and generates financial models, DCF valuations, and IC memos where every figure deep-links back to its source filing. LLMs handle judgment and narrative; deterministic Python

handles all arithmetic. ~86k lines across backend and frontend, 65 REST endpoints over 25 tables, and 437 unit tests on the financial engines.

**Tech:** React 19, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, OpenAI API, Anthropic API, SEC EDGAR, Arelle XBRL, OAuth 2.0

### **Alpha Momentum — Fintech Research Platform** · *alphamomentum.ai* · *Live*

Financial research platform I built end to end across web, iOS, and Android, with a 37k+ document SEC pipeline on AWS Batch and five LangGraph research workflows.

**Tech:** React 18, TypeScript, FastAPI, LangChain, AWS, MongoDB, Neo4j, Stripe, Plaid

### **Outlou — Location-Based Social App** · *In development*

Location-based social network in React Native and Firebase: users share moments tied to places, discover nearby activity, and connect through map-based feeds. QR profile links; English, Arabic, and French.

**Tech:** React Native, Expo, Firebase, Node.js, Google Maps API, Backblaze B2, Resend

### **EduMemes — EdTech Platform** · *edu.memes.tn* · *Live*

Educational platform that uses memes and short videos to make learning more engaging. Built solo: a React front end, a Node.js/MySQL backend, and a React Native companion app.

**Tech:** React, Node.js, Express, MySQL, React Native

## **TECHNICAL SKILLS**

**Frontend:** React 19, TypeScript, Next.js, Vite, MUI, Tailwind CSS, TanStack Query, Zustand, Framer Motion, Angular, SCSS

**Mobile:** React Native, Expo, Firebase, Plaid SDK, In-App Purchases

**Backend:** Node.js, Express, FastAPI, Flask, Django, Spring Boot, Serverless Framework

**Databases:** PostgreSQL, MongoDB, Neo4j, MySQL, DynamoDB, Firebase

**Cloud & DevOps:** AWS (Lambda, ECS, ECR, S3, CloudFront, Cognito, Batch), Docker, GitHub Actions, Nginx, Azure

**AI/ML:** LangChain, LangGraph, OpenAI API, Anthropic API, RAG, HuggingFace, TensorFlow, PyTorch

**Integrations:** Stripe, Plaid, Alpaca, OAuth 2.0, Polygon, SEC EDGAR

## **EDUCATION**

### **Computer Engineering, Data Science — ESPRIT**

*Tunis · 2021 – 2024*

- Thesis: “Generative AI with Business Knowledge” — Mention: Very Good

### **Bachelor of Computer Science — ESSECT**

*Tunis · 2018 – 2021*

- Foundation in software development, algorithms, and data structures

## **ADDITIONAL INFORMATION**

- **Languages:** Arabic (Native), English (B2, Professional), French (B2, Professional)
- **Soft skills:** Problem-solving, collaboration, and clear communication